



TRAINING PROGRAMME

Python for Data Science

Explore, query and visualise data with Python, pandas and SQL

21 HOURS

FOR Analysts, students and developers who work with data



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Overview

This programme gives participants the Python data stack used in data analysis and machine learning. They add the Python features that data code relies on, work reproducibly in Jupyter, compute with NumPy arrays, clean, reshape and combine data with pandas, query databases with SQL from Python, and communicate results with Matplotlib, Seaborn and Plotly. Every chapter ends with exercises on real datasets, and the programme closes with a final project in which each participant delivers an exploratory data analysis report as a Jupyter notebook.

At a glance

#	PHASE	FOCUS	HOURS
01	Tools & NumPy	Python for data work, Jupyter and reproducibility, NumPy arrays	6 h
02	Pandas, SQL & Visualisation	DataFrames, cleaning and reshaping, SQL queries from Python, charts	9 h
03	Final Project	An exploratory data analysis report on a real dataset	3 h
	Buffer	Catch-up, revision, extra lab time	3 h
	Total		21 h

Key facts

AUDIENCE	Analysts, students and developers who work with data
PREREQUISITES	Python basics (course: Python Programming Fundamentals, or equivalent): variables, data structures, functions. These are not taught again
DURATION	21 h: 18 h of chapters + 3 h buffer
FORMAT	Short lectures with live coding in Jupyter, each followed by exercises on real datasets
LAB ENVIRONMENT	Runs locally: one laptop per participant with Python 3, JupyterLab, pandas, NumPy, Matplotlib, Seaborn, Plotly and SQLite; a PostgreSQL database provided by the organiser; Google Colab as a fallback; no account or cost
ASSESSMENT	Exercises validated at the end of each phase and a final analysis report presented to the group; certificate of completion for participants who complete both
OUTCOME	Participants can load, clean, query, analyse and visualise a real dataset with Python, pandas and SQL, and report their findings in a reproducible notebook

PHASE 01

6 H

Tools & NumPy

Participants add the Python features that data code uses every day, work in JupyterLab and Colab in a way others can rerun, then compute with NumPy arrays through vectorisation, broadcasting and linear algebra.

Validated exercise: a notebook that simulates and summarises a dataset with vectorised NumPy code and reruns cleanly from top to bottom.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Python for Data Work	Comprehensions, lambdas, decorators, generators, type hints, dataclasses	Lecture + Lab	1.5 h
1.2	Jupyter & Reproducibility	JupyterLab, magics, nbconvert, Colab	Lecture + Lab	1.5 h
1.3	NumPy	ndarrays, vectorisation, broadcasting, linear algebra	Lecture + Lab	3 h

PHASE 02

9 H

Pandas, SQL & Visualisation

Participants load, clean, filter, group, merge and reshape data with pandas, query SQLite and PostgreSQL databases with SQL and bring the results into pandas, then choose and build the right chart with Matplotlib, Seaborn and Plotly. Validated exercise: a question about a sales dataset answered once in SQL and once in pandas, and shown in a clear chart.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Pandas	DataFrames, input and output, filtering, groupby, merge, pivot, missing values	Lecture + Lab	3 h
2.2	SQL for Data Analysis	SELECT, joins, GROUP BY, window functions; querying SQLite and PostgreSQL from Python and pandas	Lecture + Lab	3 h
2.3	Visualisation	Matplotlib, Seaborn, Plotly	Lecture + Lab	3 h

PHASE 03

3 H

Final Project

Each participant analyses a real public dataset, reusing the exercises of the phases, and presents the findings as they would to a manager who needs to decide.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Explore & Analyse	Load and clean the dataset, answer the brief's questions with SQL and pandas, build the charts	Project	1.5 h
3.2	Report & Present	Finish the notebook as a report, present the findings; questions and feedback	Project	1.5 h

FINAL PROJECT

Exploratory Data Analysis Report

A real public dataset loaded, cleaned, queried with SQL and pandas, and explored with charts, delivered as a Jupyter notebook that states the questions, the method and the findings.

STACK Python 3, JupyterLab, pandas, NumPy, SQLite or PostgreSQL, Matplotlib, Seaborn, Plotly

DELIVERABLE Jupyter notebook report, presentation of the findings